



Chapter 8 — AI & Machine Learning Components

8.1 AI Design Principles

1. **Grounded, never free-floating.** Every AI output is generated from real platform data (events, state, documents) and cites its sources.
2. **Local-first.** Ollama-served open models (Llama 3 8B / Qwen 2.5 7B) keep the platform free, offline-capable, and private. A provider interface allows hosted APIs as a config swap.
3. **Human-in-the-loop.** AI recommends; the operator decides. Every output carries an “AI-generated — verify before acting” label.
4. **Explainable pipelines.** LangGraph makes each AI workflow an explicit, inspectable graph — not a black-box prompt.

8.2 The Four AI Subsystems

Subsystem	Input	Output	Phase
Situation Reporter	Recent events + world state	Structured sitrep with citations	3
Knowledge Assistant (RAG)	Operator question + document base	Cited answer from SOPs/manuals	3
Threat Detection (CV)	Prerecorded video	Geotagged detections + risk scores	4
Predictive Analytics	Run telemetry	Forecasts with uncertainty bands	6

8.3 Situation Reporter (LLM)

LangGraph pipeline:

```
collect_events → filter_dedupe → cluster_by_topic → rank_by_severity  
→ draft_summary (LLM) → verify_grounding → format_and_store
```

- **collect_events:** pull events since last sitrep (cap ~100; prioritize critical).
- **cluster_by_topic:** group by entity/sector/kind so the LLM sees organized context, not a raw dump.
- **verify_grounding:** post-generation check — every claim in the draft must map to ≥ 1 source event ID; ungrounded sentences are stripped. This is the anti-hallucination gate for the FR-3.x acceptance criteria.

Prompt skeleton:

```
SYSTEM: You are a military-exercise situation report writer  
for a SIMULATED  
training environment. Summarize ONLY the events provided. E  
very bullet must  
end with its source event IDs in brackets, e.g. [e-48214].  
Structure:  
Summary / Key developments / Risks / Recommended actions. M  
ax 180 words.  
Never invent units, locations, or events.  
  
USER: Sim time: {sim_time}  
World snapshot: {compact_state}  
Events:  
{numbered_event_list}
```

Operator chat reuses the same grounding context plus conversation history, streamed via SSE. Questions about procedures automatically route to the RAG subsystem (a LangGraph router node decides).

8.4 Knowledge Assistant (RAG)

Corpus: self-authored SOPs (evacuation, resupply, weather protocols), simulation manuals, map/terrain notes, past incident documentation — all public or self-written.

Ingestion pipeline:

```
document → clean → chunk (500 tokens, 15% overlap, heading-aware)
→ embed (bge-base / e5-base, 768-dim) → store in pg vector (HNSW)
```

Query pipeline:

```
question → embed → top-12 vector search → keyword re-rank (hybrid)
→ top-4 chunks → LLM answer with inline [doc:chunk] citations
→ confidence check → answer OR "not found in knowledge base"
```

Key behaviors

- Refuses to answer from general model knowledge when the corpus lacks coverage — says so explicitly (trust > fluency).
- Answers show clickable source chunks in the UI side panel.
- Evaluation: a golden set of 40 Q → expected-source pairs; retrieval hit-rate and citation precision tracked per release.

8.5 Threat Detection (Computer Vision)

Pipeline detail (Celery worker):

```
video (MinIO) → ffmpeg frame sampling (2 fps)
→ YOLOv8 inference (classes: vehicle, person, smoke, fire, damage)
→ NMS + tracking (ByteTrack) to dedupe across frames
→ risk scoring → detections table → DetectionCreated events → map
```

Risk scoring formula (v1):

```
risk = w_class × confidence × (1 + 0.3 × cluster_size) × proximity_factor
```

```
w_class: fire 1.0, smoke 0.8, damage 0.7, vehicle 0.5, person 0.4
proximity_factor: 1.5 if within 2 km of any friendly unit
else 1.0
risk > 0.75 → auto-create incident (severity: warning)
```

Model strategy: start with pretrained COCO weights (vehicle/person work out of the box); fine-tune on public aerial datasets (e.g., VisDrone, xView, FireNet-class sets) for smoke/fire/damage. Training runs are MLflow-tracked; weights versioned in MinIO.

Geolocation note: since footage is prerecorded and not georeferenced, each vision job is assigned a scenario location (operator picks the sector on submit); detections spread within that zone deterministically. This is documented in-UI to keep the simulation honest.

8.6 Predictive Analytics

Feature store: per-tick telemetry aggregated to 1-min windows: movement km, active incidents, resource deltas, weather intensity per sector.

Model	Type	Features → Target	Serving
Fuel/medkit consumption	Prophet + XGBoost ensemble	usage history, active missions, weather → next-6h consumption	batch every 5 sim-min
Mission ETA	XGBoost regression	distance remaining, speed factors, weather forecast → ETA + quantile interval	on mission update
Vehicle maintenance risk	XGBoost classifier	usage hours, distance, terrain mix → failure probability	batch per hour
Supply reorder	Prophet	depot outflows → 24h reorder quantities	end of run + daily

Rules

- Every forecast stores model version + training window (`forecasts` table) — full lineage.
- Quantile outputs (P10/P50/P90) render as uncertainty bands; bare point estimates are banned in the UI (FR-10.2).

- **Cold-start:** the first runs use simple burn-rate extrapolation; ML models activate once ≥ 10 runs of telemetry exist. The dashboard labels which mode is active.

8.7 AI Safety & Quality Controls

- **Prompt-injection defense:** unit report text is untrusted; it is delimited and never interpreted as instructions.
- **Output validation:** sitreps and RAG answers are schema-checked (sections present, citations resolvable) before display; failures retry once, then surface a graceful error.
- **Latency budgets:** sitrep < 15 s; chat first-token < 2 s; RAG answer < 8 s — enforced with timeouts and fallbacks (shorter context on retry).
- **Evaluation harness:** `make eval-ai` runs the golden sets (sitrep grounding, RAG retrieval, CV mAP on a validation clip) and prints a scorecard; CI fails on regression beyond thresholds.
- **Cost:** zero — all models run locally. Hardware profile in Chapter 4 §4.6.